

Vinaykumar Venkateshkumar

linkedin.com/in/vinaykumar-venkateshkumar
github.com/Itsvid · vinaykumar.is-a.dev
Cranfield, United Kingdom

kumarvsvinay@gmail.com
+44 7742 914241

Work authorisation: sponsorship required
Available: September 2026

Design engineer with an aerospace propulsion background. Produces release-ready mechanical design work — dimensioned detail and general arrangement drawings with geometric tolerancing to ISO 1101, limits and fits to ISO 286, tolerance stack-up, sheet-metal flat patterns and design-for-manufacture checks — from parametric CAD models built in CATIA V5, CadQuery and pyOCC. Verifies geometry rather than assuming it. MSc Thermal Power and Propulsion at Cranfield. Seeking an entry-level design engineering role in aerospace or propulsion hardware.

TECHNICAL SKILLS

Mechanical Design	Geometric tolerancing (ISO 1101), limits and fits (ISO 286), tolerance stack-up (worst-case and RSS), design for manufacture, sheet-metal forming (bend allowance, flat patterns), detail and GA drawings, BOM generation
CAD	CATIA V5 (proficient), CadQuery, pyOCC / Open CASCADE, ANSYS DesignModeler, STEP and DXF interchange, parametric geometry (iCST)
CFD & Simulation	ANSYS Fluent, ANSYS ICEM CFD / Meshing, OpenFOAM, $k-\epsilon$ / $k-\omega$ turbulence modelling, grid-independence studies
Programming	Python (NumPy, SciPy, Matplotlib, pytest), MATLAB, Git, automated test suites
Engineering Domains	Gas turbine performance, propulsion, turbomachinery aerodynamics, thermodynamics, aircraft structures, non-destructive testing

DESIGN PROJECTS

- **Parametric Actuator and Gearbox Generators with Drawing Pack — (Mechanical design, GD&T, CadQuery):** Wrote two parametric generators that size a design before drawing it: a hydraulic flight-control actuator family across four aircraft classes, and an engine accessory gearbox across five power ratings (5–50 kW), with involute gear geometry, Lewis bending-stress module sizing and SKF 60xx bearing selection. Produced a four-sheet A4 drawing pack — three detail drawings and a ballooned assembly GA — carrying ISO 286 fits ($\varnothing 35$ H8 bore, $\varnothing 21$ f7 rod), ISO 1101 geometric tolerances, a chosen datum scheme, surface finishes, a parts list and a five-contributor tolerance stack (390 mm nominal, worst case ± 0.90 , RSS ± 0.44). Verification found five real defects, including a mirrored involute profile that produced an invalid solid and a housing that did not enclose its own gear. 83 automated tests. *Tech: Python, CadQuery, OpenCASCADE, ISO 286, ISO 1101, STEP*
- **Formed Sheet-Metal Equipment Bracket — (Sheet-metal design, DFM):** Parametric formed bracket with its flat pattern derived from a K-factor neutral axis, not assumed — 101.52 mm blank against 105.00 mm of summed outside legs. Validated by conservation of volume: the formed solid and the flat blank agree to 1.07%, against 5.12% for a naively summed blank. Five design-for-manufacture rules (minimum bend radius, flange length, hole-to-bend distance, fastener edge distance, minimum hole diameter) with a material trade that has a real answer: 2024-T3 is twice as strong as 5052-H32 and cannot make the fold, needing 4T minimum bend radius against 1T. Detail drawing with bend table; STEP and DXF export. *Tech: Python, CadQuery, sheet-metal forming, DFM*

PROPULSION AND ANALYSIS PROJECTS

- **Installation Aerodynamics of Aero-Engine Nacelles — Cranfield Individual Research Project (Parametric CAD, CFD):** Fitted a Class-Shape Transformation curve directly to NASA TM 110300's published NACA 1-85-100 cowl ordinates, reproducing them to 0.019% RMS of maximum radius. Built and converged a compressible RANS OpenFOAM case (rhoSimpleFoam, $k-\omega$ SST) at Mach 0.79, meshed with snappyHexMesh at 108,904 cells, and compared surface pressure against the same report's wind-tunnel data — RMS C_p error 0.65 over 17 matched stations, reported including where the model misses the leading-edge suction peak. *Tech: pyOCC, Open CASCADE, iCST, OpenFOAM, Python (Ongoing, expected Sep 2026)*
- **Twin-Spool Turbofan Cycle Model — (Gas turbine performance):** Station-by-station separate-exhaust cycle solver with every turbine pressure ratio matched to the power its own spool demands rather than assumed. 57.67 kN net thrust, 46.7% thermal efficiency at the reference design point, validated against the closed-form ideal-Brayton limit to within 0.01%. 82 tests. *Tech: Python, Matplotlib, pytest, MATLAB/Simulink cross-check*
- **Investigation of Controlled Jets for Enhanced Mixing Rates — BEng Final Year Project, team of 3 (CFD, experimental validation):** Investigated passive flow-control tab geometries (Delta Tandem Tab, M Delta Tandem Tab) for jet mixing using ANSYS Fluent across Mach 0.6–1.0, with grid-independence studies to 3.4M+ nodes, validated against experimental shadowgraph imaging. Achieved 64–76% core-length reduction against an uncontrolled baseline at Mach 0.8 with no significant thrust penalty. *Tech: ANSYS Fluent, ANSYS DesignModeler, ANSYS ICEM CFD (Mar 2025)*

Further work at vinaykumar.is-a.dev: airfoil panel method written from scratch and cross-checked against XFOil (lift

agreeing to 0.047 RMS), flight-performance calculator validated against three aircraft, OpenFOAM NACA 0012 study (GCI 6.4%), parametric wing and compressor blade-row generators.

EDUCATION

Cranfield University Cranfield, UK
MSc Thermal Power and Propulsion Oct 2025 – Sep 2026
Modules: Gas Turbine Performance, Combustion, Turbomachinery Aerodynamics, Propulsion System Design
Research project (ongoing): Installation aerodynamics of aero-engine nacelles

KCG College of Technology, Anna University Chennai, India
BEng Aeronautical Engineering — CGPA 7.37/10 2021 – 2025
Courses: Air Breathing Propulsion, Rocket Propulsion, Aero Engineering Thermodynamics, Aerodynamics I & II, Computational Fluid Dynamics, Aircraft Structures I & II, Finite Element Methods, Wind Tunnel Techniques, Non-Destructive Testing, Aircraft Design

EXPERIENCE

AI Engineering Services Limited (AIESL) — Boeing 737 MRO Base Thiruvananthapuram, India
Engineering Intern, Aircraft Maintenance Mar 2025 – Apr 2025

- Rotational internship across Component Overhaul, Material/Production Planning and Stores at a 737 base-maintenance facility servicing Air India Express.
- Documented wheel and brake overhaul procedures to specification (torque 158 lb-ft, tyre pressure 205 ± 5 psi) and Eddy Current Testing for non-destructive flaw detection.
- Used AMOS and RAMCO systems to track parts issuance, stock levels and task-card compliance.

CERTIFICATIONS AND ACHIEVEMENTS

- CATIA V5 (Proficient) — CADD Centre Training Services, 80 hrs — 2024
- IELTS Academic — Overall Band 6.5 (L 7.0, R 6.5, W 6.5, S 5.5); UKVI B2 — Jan 2025
- MATLAB Essential Training — LinkedIn Learning — 2024
- Machine Learning with Python; Deep Learning with TensorFlow — IBM / Cognitive Class — 2024